Advertising Campaign Performance Analysis

A reproducible end-to-end analysis of digital campaign efficiency, engagement, conversion, and revenue.

1. Executive Summary

The analysis covers 1,200 campaign observations. Total advertising spend is approximately $1,029,709, generating approximately $8,597,014 in tracked revenue and 73,811 conversions. Overall ROAS is 8.35, while the aggregate click-through rate is 2.15%. The results provide a practical view of channel efficiency and a reproducible basis for budget decisions.

2. Data Preparation

The workflow validates required fields, converts dates to a consistent format, checks numeric values for invalid observations, and derives CTR, conversion rate, CPC, and ROAS. The source file is retained unchanged so that every transformation can be traced to the original data.

3. Exploratory Analysis

Platform-level aggregation is used to compare spend, revenue, clicks, conversions, CTR, conversion rate, and ROAS. Monthly aggregation is used to identify revenue movement through the year. The supplied charts are generated by the R scripts and can be regenerated from the source dataset.

Platform	Spend	Revenue	ROAS	CTR	CVR
Meta Ads	$343,288	$3,532,372	10.29	2.19%	6.40%
Google Ads	$364,210	$3,445,409	9.46	3.42%	9.35%
YouTube Ads	$149,373	$1,177,276	7.88	1.51%	3.94%
LinkedIn Ads	$172,838	$441,957	2.56	1.70%	5.13%

4. Predictive Modeling

A linear regression model estimates revenue from spend, impressions, clicks, conversions, platform, campaign type, and audience. An 80/20 train-test split and fixed random seed make the evaluation reproducible. RMSE, MAE, and test-set R^2 are reported in the generated model metrics file. The model is intended as an analytical benchmark rather than a production forecasting system.

5. Interpretation and Recommendations

Budget decisions should consider ROAS together with scale and conversion quality rather than a single metric. High-ROAS channels should be reviewed for additional scalable budget, while lower-performing channels should be tested through audience, creative, bidding, or placement changes before being expanded. Monthly trends should be monitored to identify periods where spend and revenue diverge.

6. Limitations

The dataset is structured for reproducible analytical practice and does not represent a live advertising account. Attribution quality, customer lifetime value, creative-level effects, seasonality controls, and external market conditions are not modeled. Therefore, findings should be treated as analytical evidence within the dataset rather than causal claims.

7. Reproducibility

Run `run_all.R` from the repository root. It validates the data, regenerates summaries and plots, fits the model, and writes the resulting metrics and insight tables to `outputs/`.